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> # Create the dataset manually
> data <- data.frame(
+   Sector = c("Residential", "Commercial", "Industrial", "Residential", "Commercial", "Industrial"),
+   Region = c("North", "South", "West", "East", "North", "South"),
+   Month = c("Jan", "Jan", "Feb", "Feb", "Mar", "Mar"),
+   Temperature = c(15, 24, 20, 18, 28, 30),
+   Units_Consumed = c(320, 540, 880, 350, 610, 920),
+   Cost = c(2100, 3600, 5900, 2300, 4100, 6200),
+   Renewable_Usage = c(22, 18, 12, 25, 20, 15),
+   Peak_Hours = c(4, 6, 8, 5, 7, 9)
+ )
>
> print(data)
  Sector Region Month Temperature Units_Consumed Cost Renewable_Usage
1 Residential North  Jan         15           320 2100             22
2 Commercial South  Jan         24           540 3600             18
3 Industrial  West  Feb         20           880 5900             12
4 Residential East   Feb         18           350 2300             25
5 Commercial North  Mar         28           610 4100             20
6 Industrial South  Mar         30           920 6200             15
Peak_Hours
1      4
2      6
3      8
4      5
5      7
6      9
>
> library(ggplot2)
Need help? Try Stackoverflow: https://stackoverflow.com/tags/ggplot2
>
> # Helper function to open a new platform-appropriate window
> new_window <- function() {
+   if (.Platform$OS.type == "windows") {
+     windows()
+   } else {
+     quartz()
+   }
+ }
>
> # PLOT 1: Histogram
> new_window()
> hist(data$Units_Consumed,
+   col = "skyblue",
+   main = "Histogram of Units Consumed",
+   xlab = "Units Consumed (kWh)",
+   ylab = "Frequency",
+   border = "black")
>
> # PLOT 2: Density Plot
> new_window()
> plot(density(data$Units_Consumed),
+   main = "Density Plot of Units Consumed",
```

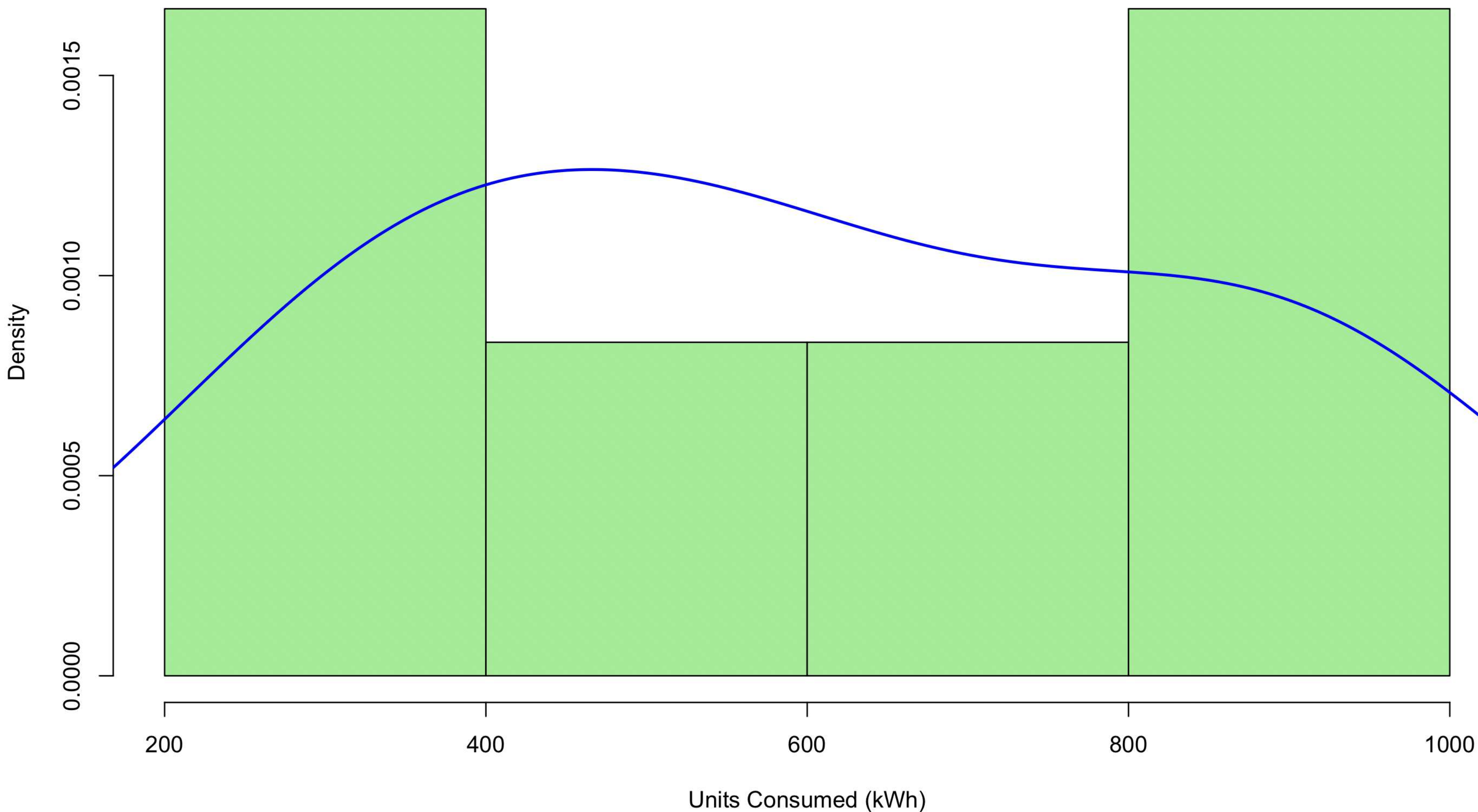


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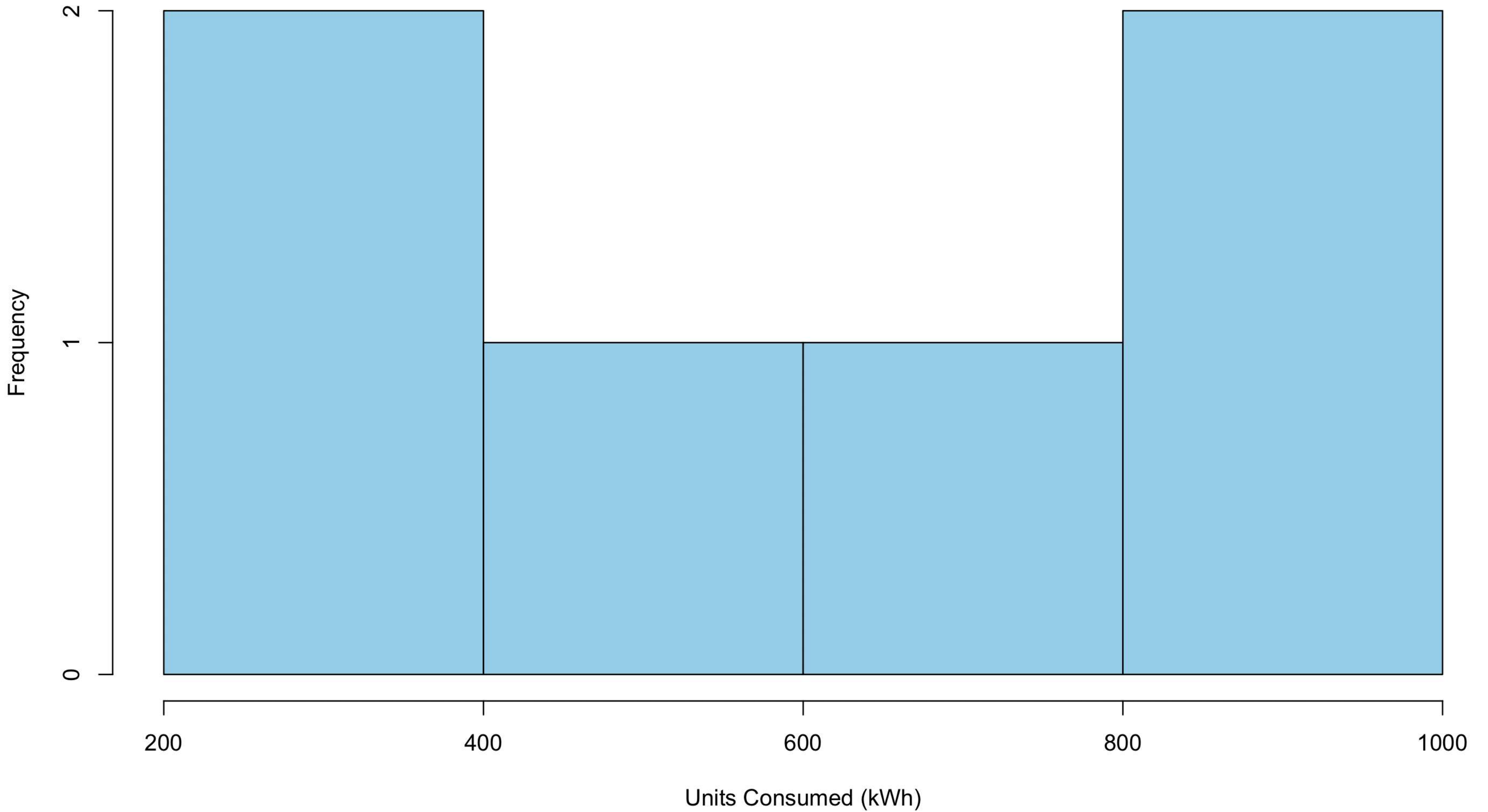
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```
> # PLOT 1: Histogram
> new_window()
> hist(data$Units_Consumed,
+       col = "skyblue",
+       main = "Histogram of Units Consumed",
+       xlab = "Units Consumed (kWh)",
+       ylab = "Frequency",
+       border = "black")
>
> # PLOT 2: Density Plot
> new_window()
> plot(density(data$Units_Consumed),
+       main = "Density Plot of Units Consumed",
+       xlab = "Units Consumed (kWh)",
+       col = "darkred",
+       lwd = 2)
>
> # PLOT 3: Histogram + Density Combined
> new_window()
> hist(data$Units_Consumed, probability = TRUE, col = "lightgreen",
+       main = "Histogram with Density Curve", xlab = "Units Consumed (kWh)")
> lines(density(data$Units_Consumed), col = "blue", lwd = 2)
>
> # PLOT 4: Scatter/Bubble Plot
> new_window()
> print(
+   ggplot(data, aes(x = Temperature, y = Units_Consumed, size = Peak_Hours)) +
+   geom_point(alpha = 0.5, color = "purple") +
+   scale_size(range = c(3, 12)) +
+   labs(title = "Temperature vs Units Consumed (Bubble Size = Peak Hours)",
+        x = "Temperature (°C)",
+        y = "Units Consumed (kWh)",
+        size = "Peak Hours") +
+   theme_minimal()
+ )
>
> # PLOT 5: Bar Chart
> avg_renewable <- aggregate(Renewable_Usage ~ Sector, data = data, FUN = mean)
> print(avg_renewable)
  Sector Renewable_Usage
1 Commercial          19.0
2 Industrial          13.5
3 Residential          23.5
>
> new_window()
> print(
+   ggplot(avg_renewable, aes(x = Sector, y = Renewable_Usage, fill = Sector)) +
+   geom_bar(stat = "identity", width = 0.5) +
+   geom_text(aes(label = round(Renewable_Usage, 1)), vjust = -0.5) +
+   labs(title = "Average Renewable Energy Usage by Sector",
+        x = "Sector", y = "Average Renewable Usage (%)") +
+   theme_minimal()
+ )
>
> |
```

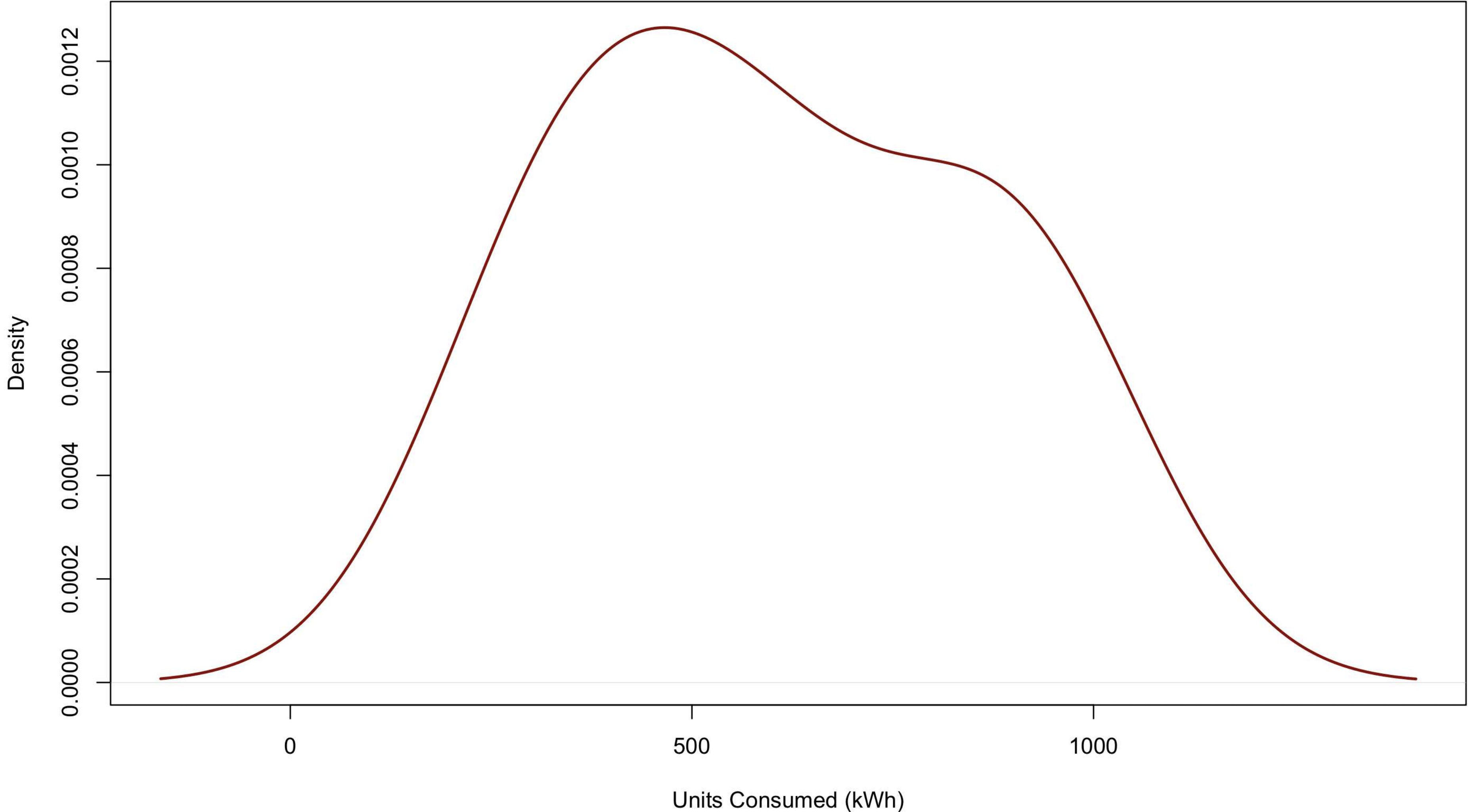
Histogram with Density Curve



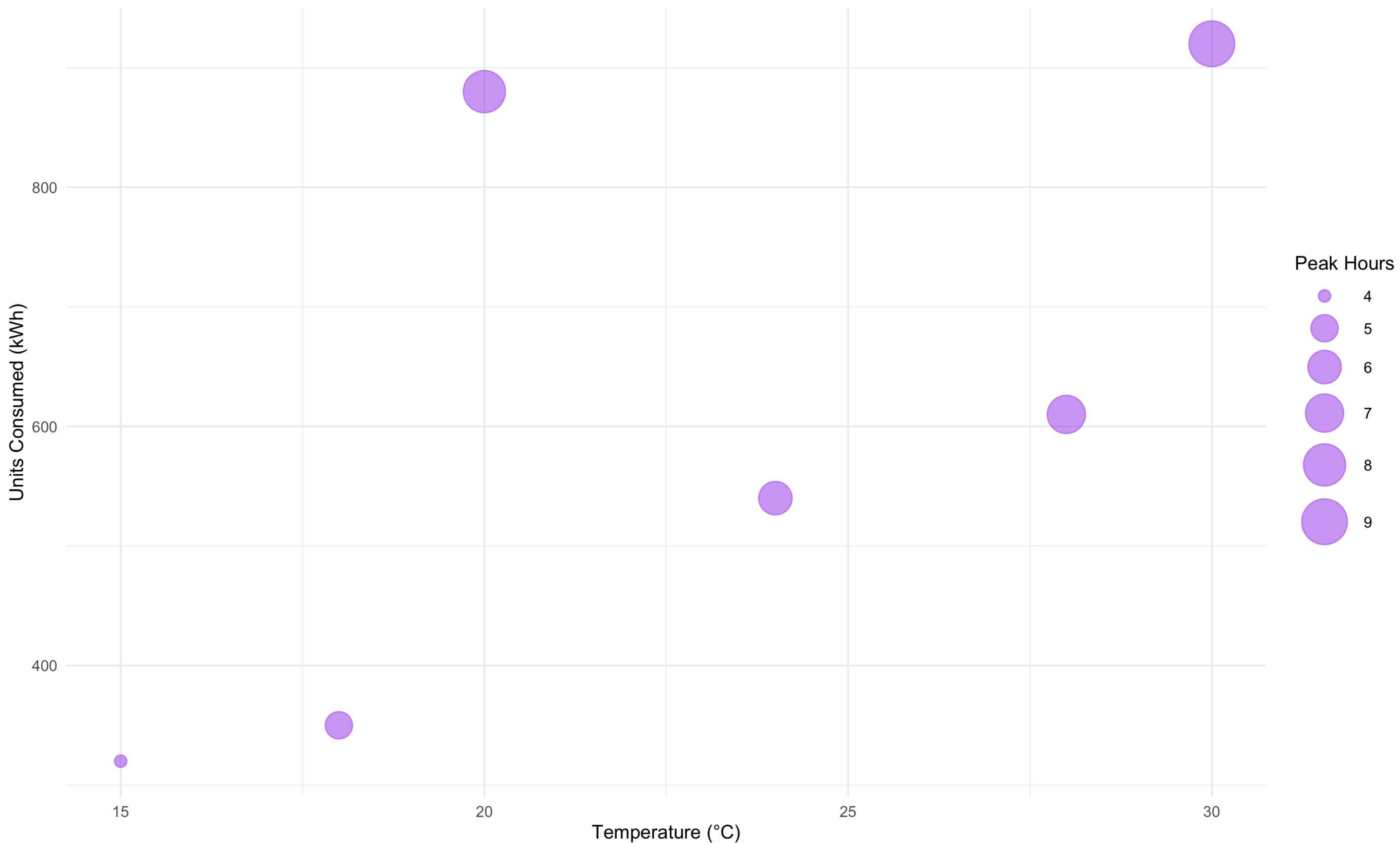
Histogram of Units Consumed



Density Plot of Units Consumed



Temperature vs Units Consumed (Bubble Size = Peak Hours)



Average Renewable Energy Usage by Sector

